Cross-Modality Neonatal Brain Image Conversion Using a Latent Diffusion Model

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I. Abstract

Multi-contrast neonatal magnetic resonance imaging (MRI) is invaluable because T1-weighted (T1w) and T2-weighted (T2w) images provide complementary information about tissue properties. However, time constraints and the need for sedation often preclude the acquisition of both modalities. Here, we propose a 2D slice-wise conditional latent diffusion model (LDM) that synthesizes T1w images from routinely collected T2w scans in neonates. Forty bias field-corrected T1w and T2w pairs were sliced along the z-axis and encoded using an Autoencoder-KL. Then, synthetic T1w images were generated via conditional image-to-image translation by injecting T2w latents alongside modality-specific conditioning information. On a held-out test set, the model achieved a peak signal-to-noise ratio of 21.03 dB and a structural similarity index matrix of 0.6963. This cross-modality conversion using LDM enables T1w image generation from T2w alone, providing a fast and sedation-free solution for multi-contrast neonatal MRI.

II. Introduction

Magnetic resonance imaging (MRI) is a non-invasive technique for visualizing brain structure. Multimodal MRI provides rich information to identify specific brain patterns in both healthy controls and patients with neurological disorders. T1-weighted (T1w) and T2-weighted (T2w) scans, in particular, provide complementary information about brain structure. T1w emphasizes myelinated white matter and acute hemorrhage, whereas T2w highlights unmyelinated tissue and fluid accumulation [1]. Additionally, the T1w/T2w intensity ratio has been used to observe cortical microstructural profiles [2], aiding clinicians in diagnosing neurological disorders with greater efficiency. However, acquiring both contrasts in neonates poses significant challenges: scan time is strictly limited, motion artifacts are frequent, and pharmacological sedation introduces substantial risk and cost [3]. Thus, the absence of either modality can undermine the diagnostic accuracy of multiple diseases.

One way to overcome this resource shortage is a cross-modality synthesis that generates the missing modality from routinely acquired data. Early studies utilized generative adversarial networks (GANs) for this task, but the GAN-based models often suffer from mode collapse, local minima, non-convergence, and instability, due to the adversarial training process [4, 5]. On the other hand, denoising-diffusion probabilistic models (DDPMs) [6] have demonstrated superior performance and stability in image synthesis tasks [7]. These models learn a stochastic process, progressively adding noise to an image and then reversing this process using Markov chain-based inference to reconstruct the original data. It enabled the diffusion-based model to generate realistic synthetic images without relying on priors. However, these models are computationally intensive, particularly for 3-dimensional (3D) MRI data, due to the high memory requirements of repeated denoising steps. Latent diffusion models (LDMs) address this issue by operating in a compressed latent space [8]. In brief, images are encoded into a low-dimensional latent representation, and then decoded back into the original image space. Conditioning inputs can be added during the diffusion process to guide the transformation from one modality to another. Recent LDM variants, such as *Make-A-Volume* [9] and *CoLa-Diff* [10], have achieved promising results in adult multi-modal MRI synthesis. However, the application of these methods to neonatal brain imaging remains largely unexplored. Notably, the neonatal brain differs significantly from the adult brain in tissue contrasts [11], highlighting the need for neonatal-specific synthesis models [13].

In this study, we propose a 2D slice-based conditional LDM that synthesizes T1w images from T2w scans in neonates. Our pipeline offers a practical solution to generate multi-contrast neonatal brain images in resource-constrained clinical environments.

III. Methods

A. Participants and imaging data

We obtained imaging data of 40 neonates from the first release of the Developing Human Connectome Project (dHCP) database [14]. All participants were healthy term-born infants (37–44 weeks of age) with no major brain abnormalities. T1w and T2w data were acquired at the Evelina Neonatal Imaging Centre, London, using a 3T Philips Achieva scanner with a dedicated 32-channel neonatal head coil. T1w data were acquired using an inversion recovery fast spin-echo sequence (repetition time [TR] = 4,795 ms; inversion time [TI] = 1,740 ms; and echo time [TE] = 8.7 ms) with 0.8×0.8 mm in-plane resolution and 1.6 mm slice thickness. T2w images were obtained using a fast spin-echo sequence (TR = 12,000 ms and TE = 156 ms) with the same resolution. Preprocessing steps included N4 bias field correction, skull-stripping, and tissue segmentation. All 3D volumes (290×290×203 voxels) were sliced along the z-axis and grouped into triplets to form RGB-style tensors, yielding 2,736 samples per modality. The dataset was randomly partitioned into training (1,936), validation (415), and test (416) sets (i.e., 70:15:15 split).

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B. Autoencoder

LDM was applied to generate synthetic images through three main stages: compression, diffusion, and reconstruction. First, to learn compact and perceptually meaningful image representations, we trained an autoencoder with Kullback-Leibler regularization (AE-KL), which consists of a convolutional layer-based encoder–decoder structure [15]. The encoder compressed the T1w image (3×256×256) into a latent tensor (3×64×64), and the decoder reconstructed the original image from the latent representation. The model was trained to minimize the reconstruction loss between the input image x and its reconstruction $\hat{x}$. Additionally, AE-KL incorporates adversarial loss and KL loss [15]. To facilitate efficient learning, the latent vector z was regularized to be zero-centered with low variance using a KL regularization, allowing recovery into pixel space with minimal distortion. AE-KL was trained in an adversarial manner, similar to GAN [6]. The total loss function (L_{AE-KL}) used for training is as follows:

$$L_{AE-KL} = \min_{\mathcal{E},\mathcal{D}} \max_{\psi} \Big(L_{rec}\Big(x, \mathcal{D}\big(\mathcal{E}(x)\big)\Big) - L_{adv}\Big(\mathcal{D}\big(\mathcal{E}(x)\big)\Big) + \log D_{\psi}(x) + L_{reg}(x;\, \mathcal{E}, \mathcal{D}) \Big), \quad (1)$$

where $\mathcal{D}$ and $\mathcal{E}$ denote decoder and encoder, L_{rec}, L_{adv} and L_{reg} indicate reconstruction error, adversarial loss, and KL loss, and D_{ψ} is a patch-based discriminator. The autoencoder for T1w images was trained for 200 epochs with the β hyperparameter controlling the KL divergence penalty set to 1e-6. The best model was selected based on the lowest total reconstruction loss on the validation data. Similarly, AE-KL compression was also applied to T2w images, which were used as conditioning inputs during the diffusion process. It was trained for 200 epochs with $\beta = 1 \times 10^{-6}$.

C. Conditional latent diffusion model

To generate images of different modality, we trained a conditional LDM. Specifically, it synthesizes T1w latents from T2w latents using a concatenation method. Using the frozen encoder and decoder from AE-KL, the LDM operates in the latent space. During the forward process, Gaussian noise was iteratively added to the initial T1w latent z_0 over 1,000 steps, resulting in a noisy latent representation z_t ($z \in \mathbb{R}^3 \times 64 \times 64$). Diffusion process can be expressed as:

$$q(z_t|z_0) = \mathcal{N}(z_t|\, \alpha_t z_0, \sigma_t^2 \mathbb{I}), \quad (2)$$

where the forward diffusion process q starts from a latent vector z_0. Denoising diffusion formula are follows as:

$$p(z_0) = \int_z p(z_T) \prod_{t=1}^{T} p(z_t - 1|z_t), \quad (3)$$

where $p(z_{t-1}|z_t)$ can be represented below:

$$p(z_{t-1}\,|z_t) := q\big(z_{t-1}\,\big|\, z_t, z_\theta(z_t, t)\big) = \mathcal{N}(z_{t-1}\,|\mu_\theta(z_t, t), \sigma_{t|t-1}^2 \frac{\sigma_{t-1}^2}{\sigma_t^2} \mathbb{I}). \quad (4)$$

The reverse denoising process was learned using a time-conditional U-Net architecture. For paired image-to-image translation, we opted for a concatenation-based conditioning method using T2w latents y . The T2w images were compressed into latent tensors of shape $\mathbb{R}^3 \times 64 \times 64$, identical to the T1w latents. During training, we used T1w-T2w pairs and concatenate both latent tensors, with T1w latents injected as a condition [16]. Considering the latent space, the conditioning key y and diffusion time steps t, the noise prediction loss (L_{LDM}) was defined as follows:

$$L_{LDM} = \mathbb{E}_{z_0, y, \epsilon, t}[||\epsilon - \epsilon_\theta(z_t, t, y)||_2^2], \quad (5)$$

where z_t is the noisy latent at timestep t, $\epsilon \sim \mathcal{N}(0,1)$ is sampled noise, and ϵ_θ denotes the U-Net-based denoising function conditioned on y. Training lasted for 200 epochs with early stopping based on the validation loss.

D. Optimization strategy

This framework consisted of two stages: AE-KL and conditional LDM. After resizing the T1w and T2w images to $265 \times 265 \times 3$, each image was trained for 200 epochs. They shared the same optimizer parameters ($\beta_1 = 0.5, \beta_2 = 0.9$ and a learning rate of 4.5×10^{-6}). Then, the LDM was trained to generate T1w latent representations, with T2w latents used as the conditioning input. The T2w latent tensors were concatenated to provide the conditioning key for the U-Net. It was trained for 200 epochs using the Adam optimizer ($\beta_1 = 0.5, \beta_2 = 0.9$ and a learning rate of 1×10^{-6}), with the batch size of 12. For T2w-to-T1w conversion, the T1w tensors were sampled from the distribution using Denoising Diffusion Implicit Models (DDIMs), which had different conditions, transforming denoising steps with a strength parameter and guidance scale to compare the performance metrics and image quality in each case.

E. Evaluation metrics

The final model was evaluated on the test data using the peak signal-to-noise ratio (PSNR), structural similarity index measure (SSIM), mean absolute error (MAE), and normalized cross-correlation (NCC) between the actual and synthesized T1w.

IV. Results

A. Synthesis performance

By applying the proposed model to T2w scans of neonates, we successfully synthesized corresponding T1w images. To determine the optimal conditions for T2w-to-T1w modality conversion, we performed a sensitivity analysis using different combinations of hyperparameters, such as guidance scale and denoising strength, while fixing the number of DDIM steps at 100. In addition to evaluating quantitative metrics, we visually inspected the synthetic T1w images decoded and sampled from the distribution. Based on the ablation studies, the model achieved a peak PSNR of 21.03 dB, SSIM of 0.6988, MAE of 70.93, and NCC of 0.4754 (Table I). In particular, increasing the denoising strength to 90% generally enhanced structural similarity but slightly reduced PSNR. The synthetic T1w images preserved key structural features. Specifically, the synthetic images retained clear delineation of the gray-white matter boundary and exhibited enhanced visibility of detailed cortical folding patterns, particularly under higher denoising strength. Overall, the synthetic T1w images resembled the target domain, demonstrating effective and reliable modality conversion capabilities.

TABLE I. MODEL PERFORMANCE ON THE TEST SET

Scale, Strength (%)	PSNR (dB)	SSIM	MAE	NCC
3.0, 50%	21.03	0.6963	73.40	0.4713
3.0, 70%	21.02	0.6966	72.95	0.4748
3.0, 80%	20.99	0.6976	72.39	0.4725
3.0, 90%	20.91	0.6985	71.28	0.4678
5.0, 50%	21.02	0.6961	73.45	0.4715
5.0, 70%	21.03	0.6967	72.92	0.4752
5.0, 80%	20.97	0.6976	72.48	0.4714
5.0, 90%	20.92	0.6988	70.93	0.4677
7.5, 50%	21.03	0.6961	73.43	0.4718
7.5, 70%	21.02	0.6966	72.96	0.4754
7.5, 80%	20.97	0.6975	72.29	0.4728
7.5, 90%	20.93	0.6985	70.94	0.4681

Abbreviations: PSNR, peak signal-to-noise ratio; SSIM, structural similarity index measure; MAE, mean absolute error; NCC, normalized cross-correlation.

By visual inspection, the synthesized T1w images from T2w inputs showed comparable tissue boundary shapes to the ground truth (Fig. 1). Each image in the middle of Fig. 1 represents the results across different hyperparameter settings for the denoising stage. Moving to the right in Fig. 1 corresponds to increasing intensity levels, while moving downward indicates improved structural clarity. The visibility of brain structures varied depending on the hyperparameter configurations used during inference. Although details of gray matter curvature still require refinement, the synthesized outputs were highly comparable to the original T1w images.

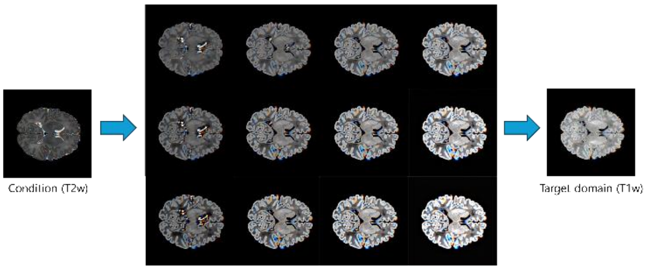

Fig. 1. Conditioning image (left), synthetic T1w images with various hyperparameter settings (middle), and target T1w image. The middle column shows the effects of different hyperparameters. The visual inspection was assessed across the overall brain structure, with particular attention to tissue contrast under different hyperparameter configurations.

V. Discussion

In this study, we proposed a 2D slice-based LDM for generating T1w images from T2w scans in neonates. To our knowledge, this is one of the earliest applications of LDM for neonatal cross-modality MRI synthesis, filling a gap left by prior adult-focused studies. By synthesizing the missing T1w contrast from a single T2w scan, our approach can help reduce scan time, avoid the need for pharmacological sedation, and still provide critical tissue contrast information necessary for early diagnosis of various neurological conditions.

However, several limitations should be acknowledged. The use of 2D-based model may result in discontinuities along the z-axis, warranting future investigation using 3D perceptual modeling. Additionally, the small sample size may limit the generalizability of the model. This issue can be addressed via data augmentation or the incorporation of external datasets. Moreover, we will test the robustness of the model to fetal MRI, which is also difficult to obtain the data.

Ultimately, our work may provide clinicians with sufficient information to support accurate diagnosis of neurodevelopmental disorders [18]. These directions aim to advance our model toward practical deployment in neonatal and broader pediatric neuroimaging settings.

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